

ASSIGNMENT #4

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: _____ ESTEBAN VAZQUEZ-ELIZONDO _____
Due Date: Friday March 7 by 5:00pm in my office

Number of points.....100
Your score.....

Note: All problems are worth 100 points. Your final score will be the average value.

Problem 4A

The first problem is to determine the Laplace transform of an input signal based on a graphical representation. Figure 1 is the graph that was given:

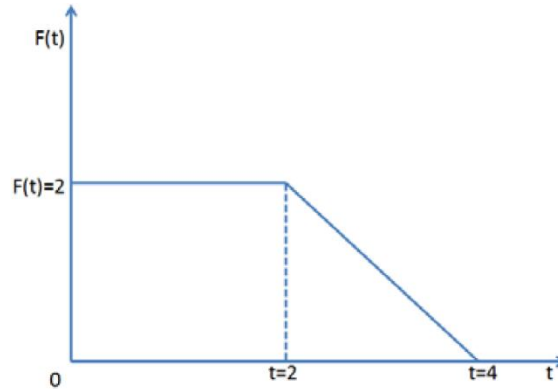


Figure 1 – Given Graph for Problem 4A

To start the problem, we can determine the bounds as:

$$F(t) = \begin{cases} 2 & 0 \leq t \leq 2 \\ 4 - t & 2 \leq t \leq 4 \end{cases} \quad (1)$$

From Equation 1, we are ready to take the Laplace transform as follows:

$$\mathcal{L}[F(t)] = \int_0^2 2e^{-st} dt + \int_2^4 (4 - t)e^{-st} dt \quad (2)$$

The first section of the integral from 0 to 2 is a straight-forward calculation. However, the second section of limits 2 to 4 will require integration by parts. Equation 2 can then be solved as follows:

$$\mathcal{L}[F(t)] = -\frac{2e^{-st}}{s} \Big|_0^2 + (4 - t) \int_2^4 e^{-st} dt - \int_2^4 \left(\frac{d}{dt}(4 - t) \right) e^{-st} dt$$

$$\mathcal{L}[F(t)] = -\frac{2(e^{-2s} - 1)}{s} - \frac{(4 - t)e^{-st}}{s} \Big|_2^4 + \frac{1}{s^2} e^{-st} \Big|_2^4$$

$$\mathcal{L}[F(t)] = \frac{2}{s} - \frac{2e^{-2s}}{s} + \frac{2e^{-2s}}{s} + \frac{e^{-4s}}{s^2} - \frac{e^{-2s}}{s^2} = \frac{2}{s} + \frac{e^{-4s}}{s^2} - \frac{e^{-2s}}{s^2}$$

$$\therefore \mathcal{L}[F(t)] = \frac{2}{s} + \frac{e^{-4s}}{s^2} - \frac{e^{-2s}}{s^2}$$

Problem 4B

The following equations were given to find the inverse of the Laplace transforms using partial fraction expansions.

$$Y_1(s) = \frac{s + 4}{(s + 1)^2(s + 2)} \quad (3)$$

$$Y_2(s) = \frac{8(s + 2)}{s(s^2 + 4)} \quad (4)$$

Using the Heaviside method, we can solve Equation 3 as follows:

$$Y_1(s) = \frac{s + 4}{(s + 1)^2(s + 2)} = \frac{A}{(s + 1)^2} + \frac{B}{(s + 1)} + \frac{C}{(s + 2)} \quad (5)$$

To solve for A, we need to multiply by denominator of the fraction of A. Both sides are multiplied by $(s+1)^2$ and the value of s is set to zero out the function so $s = -1$:

$$Y_1(-1) = \frac{-1 + 4}{(-1 + 2)} = A + \frac{(-1 + 2)^2 B}{(-1 + 1)} + \frac{(-1 + 2)^2 C}{(-1 + 2)}$$

$$A = 3$$

To solve for C, similar steps as A. We zero out the denominator on the right side, so we set $s = -2$. We then multiply both sides by $(s+2)$:

$$Y_1(-2) = \frac{-2 + 4}{(-2 + 1)^2} = \frac{(-2 + 2)A}{(-2 + 1)^2} + \frac{(-2 + 2)B}{(-2 + 1)} + C$$

$$C = 2$$

To solve for B, an arbitrary value is chosen for s, I will pick $s = 0$. The values of A and C are known so we can plug those and multiply both sides by $(s+1)$:

$$Y_1(0) = \frac{0 + 4}{(0 + 1)(0 + 2)} = \frac{(0 + 1)A}{(0 + 1)^2} + \frac{(0 + 1)B}{(0 + 1)} + \frac{(0 + 1)C}{(0 + 2)}$$

$$Y_1(0) = 2 = \frac{3}{1} + B + \frac{2}{2} \rightarrow B = -2$$

Next, we plug the values of A, B, and C back into Equation 5 and take inverse Laplace:

$$\frac{s + 4}{(s + 1)^2(s + 2)} = \frac{3}{(s + 1)^2} + \frac{-2}{(s + 1)} + \frac{2}{(s + 2)}$$

$$\mathcal{L}^{-1} \left[\frac{3}{(s + 1)^2} \right] = 3te^{-t}$$

$$\mathcal{L}^{-1} \left[\frac{-2}{(s + 1)} \right] = -2e^{-t}$$

$$\mathcal{L}^{-1} \left[\frac{2}{(s + 2)} \right] = 2e^{-2t}$$

$$\therefore \boxed{\mathcal{L}^{-1}[Y_1(s)] = 3te^{-t} - 2e^{-t} + 2e^{-2t}}$$

Problem 4B (Continued)

We will now solve for Equation 4 using the Heaviside method as well:

$$Y_2(s) = \frac{8(s+2)}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{(s^2+4)} \quad (6)$$

To solve for A, we multiply Equation 6 by s on both sides and set s = 0:

$$Y_2(0) = \frac{8(0+2)}{(0^2+4)} = A + \frac{B(0)^2}{(0^2+4)} + \frac{C(0)}{(0^2+4)}$$

$$A = 4$$

Next, we solve for B and C by multiplying the Heaviside formula by (s²+4) and setting s = -2:

$$Y_2(-2) = \frac{8(-2+2)}{-2} = \frac{A(-2^2+4)}{-2} + B(-2) + C$$

$$Y_2(-2) = 0 = -4A - 2B + C \quad (7)$$

We can plug value of A in Equation 7:

$$Y_2(-2) = 0 = -16 - 2B + C$$

$$Y_2(-2) = 16 = C - 2B \quad (8)$$

We need to find another formula that relates B and C besides Equation 8. This can be done by multiplying s and (s²+4) in Equation 6:

$$Y_2(s) = 8(s+2) = A(s^2+4) + Bs^2 + Cs$$

After obtaining this expression, we can take the derivative and set s = 0 to obtain a value:

$$Y_2(s) = 8 = 2A(0) + 2B(0) + C$$

$$C = 8$$

With the value of C, we can plug into Equation 8 to obtain value of B:

$$Y_2(-2) = 8 - 16 = 2B \rightarrow -8 = 2B$$

$$B = -4$$

Finally, we plug the values of A, B, and C back into Equation 6 and take inverse Laplace:

$$\frac{8(s+2)}{s(s^2+4)} = \frac{4}{s} + \frac{-4s}{(s^2+4)} + \frac{8}{(s^2+4)}$$

$$\mathcal{L}^{-1} \left[\frac{4}{s} \right] = 4$$

$$\mathcal{L}^{-1} \left[\frac{-4s}{(s^2+4)} \right] = -4\cos(2t)$$

$$\mathcal{L}^{-1} \left[\frac{8}{(s^2+4)} \right] = 8\sin(2t)$$

$$\therefore \mathcal{L}^{-1}[Y_2(s)] = 4 - 4\cos(2t) + 8\sin(2t)$$

Problem 4C

Find solution using Laplace transforms and plot solution from {t=0 to t=2}, given:

$$\frac{du}{dt} + 4u = f(t) \quad (9)$$

For t = 0 and u = 0 then:

$$f(t) = \begin{cases} 0 & t < 0 \\ 5 & 0 \leq t \leq 0.2 \\ 0 & 0.2 \leq t \leq 2 \end{cases}$$

Take the Laplace of Equation 9 to find $G_p(s)$ by using $u(s)=G_p(s)f(s)$ form:

$$\begin{aligned} \mathcal{L}\left[\frac{du}{dt} + 4u = f(t)\right] \\ su(s) + 4u(s) = f(s) \\ u(s)(s + 4) = f(s) \\ G_p(s) = \frac{1}{s + 4} \end{aligned}$$

With the given bounds, we can derive f(s) as:

$$\begin{aligned} f(s) &= \int_0^\infty f(t)e^{-st}dt = \int_{-\infty}^0 0e^{-st}dt + \int_0^{0.2} 5e^{-st}dt + \int_{0.2}^2 0e^{-st}dt \\ f(s) &= 5 \int_0^{0.2} e^{-st}dt = 5 \left[-\frac{1}{s}e^{-st} \right]_0^{0.2} = -\frac{5}{s}e^{-0.2s} + \frac{5}{s}e^{-0s} = -\frac{5}{s}(e^{-0.2s} - 1) \end{aligned}$$

Then u(s) is defined as the following:

$$u(s) = \frac{1}{s + 4} \times -\frac{5}{s}(e^{-0.2s} - 1) = -\frac{5(e^{-0.2s} - 1)}{s(s + 4)} = -5 \left[\frac{e^{-0.2s}}{s(s + 4)} - \frac{1}{s(s + 4)} \right]$$

Next, we use the Heaviside method on the second section of u(s):

$$\frac{1}{s(s + 4)} = \frac{A}{s} + \frac{B}{(s + 4)} \quad (11)$$

Multiply Equation 11 by s and set s = 0:

$$\begin{aligned} \frac{1}{(0 + 4)} &= A + \frac{B(0)}{(0 + 4)} \\ A &= \frac{1}{4} \end{aligned}$$

Next, we multiply Equation 11 by (s+4) and set s = -4:

$$\begin{aligned} \frac{1}{-4} &= \frac{A(-4 + 4)}{-4} + B \\ B &= -\frac{1}{4} \end{aligned}$$

Plug the values of A and B back into Equation 11:

$$\frac{1}{s(s + 4)} = \frac{1}{4s} - \frac{1}{4(s + 4)}$$

Problem 4C (Continued)

We can take the Laplace inverse of $u(s)$ and apply time delay $t-0.2$ from t :

$$\mathcal{L}^{-1}[u(s)] = -5 \left[\frac{1}{4} - \frac{1}{4} e^{-4(t-0.2)} - \frac{1}{4} + \frac{1}{4} e^{-4t} \right]$$

$$u(t) = \frac{5}{4} e^{-4(t-0.2)} - \frac{5}{4} e^{-4t}$$

$$\therefore u(t) = \frac{5}{4} [e^{-4(t-0.2)} - e^{-4t}]$$

Next, we can plot $u(t)$ by using the boundaries we were initially given:

$$u(t) = \begin{cases} \frac{5}{4} [e^{-4(t-0.2)} - e^{-4t}] & 0 \leq t \leq 0.2 \\ 0 & 0.2 < t \leq 2 \end{cases}$$

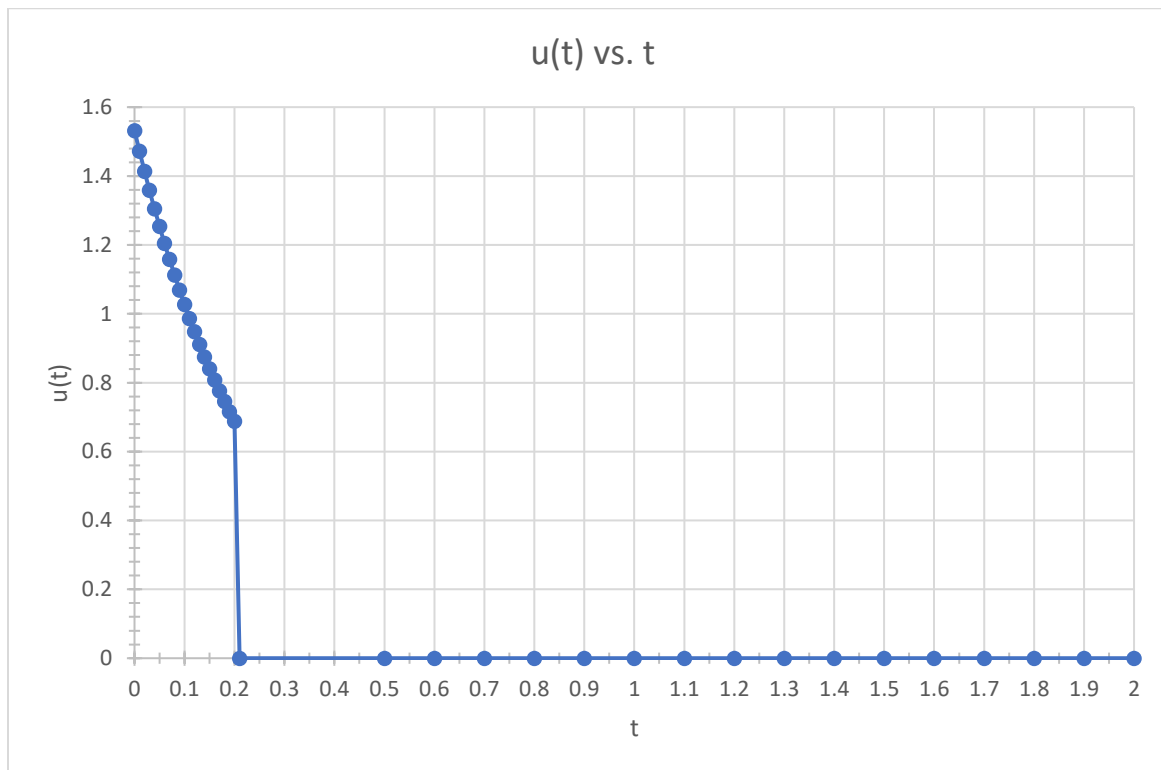


Figure 2 – Plot of $u(t)$ vs. t from $t = 0$ to $t = 2$

Problem 4D

Using the Fundamental theorem on functions $u(s)$, what could be said regarding $0 \leq t \leq \infty$ for the following equations are:

$$u(s) = \frac{s + 1}{s(s + 2)(s + 3)} \quad (12)$$

Equation 12 is **converging** as the roots are negative (0, -2, -3) and it is also **smooth** as there will be no imaginary numbers. The initial value of $u(0)$ will be $u(0) = \frac{1}{6}$ and the final asymptotic value $v(\infty)$ will be $u(\infty) = \frac{\infty}{\infty}$ Use L'Hopital Rule:

$$\lim_{s \rightarrow \infty} \frac{1}{2s + 6} = 0$$

Then $v(\infty) = 0$

$$u(s) = \frac{s + 4}{s(s + 2)(s^2 - 2s + 5)} \quad (13)$$

On the other hand, Equation 13 is **diverging** as there is a positive root in the third term from the quadratic formula ($s^2 - 2s + 5$). Since there is a presence of an imaginary number, then the curve is **oscillating** as well. The initial value of $u(0)$ will be $u(0) = \frac{2}{5}$ and the final asymptotic value $v(\infty)$ will be $u(\infty) = \frac{\infty}{\infty}$ Use L'Hopital Rule:

$$\lim_{s \rightarrow \infty} \frac{1}{3s^2 + 1} = 0$$

Then $v(\infty) = 0$

Functions $u(s)$	Converging/Diverging	Smooth/Oscillating	$u(0)$	$u(\infty)$
$u(s) = \frac{s + 1}{s(s + 2)(s + 3)}$	Converging	Smooth	$\frac{1}{6}$	0
$u(s) = \frac{s + 4}{s(s + 2)(s^2 - 2s + 5)}$	Diverging	Oscillating	$\frac{2}{5}$	0